

Foundations of Deep Learning

Project Proposal

Implementation of the Fundamental Components Pivotal to the Realization of Artificial General Intelligence (AGI) through the Utilization of Deep Learning Methodologies, including 20-Way One-Shot Classification, on Omniglot Dataset

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Project Idea :

The primary objective of this project is to undertake the implementation of fundamental components essential to the realization of Artificial General Intelligence (AGI) through the employing of deep learning techniques. These foundational components can be categorized as *causality*, *compositionality*, and *learning-to-learn*.

These aforementioned concepts were researched and implemented in models, particularly within the domain of Cognitive Science research^{1,2,3}. The main aim of such research can be briefly summarized as the mimic of human learning, perception and cognitive processes for machine systems : as Lake et al. (2017) put forward , *building machines that see, learn and think like people*. Hence, my approach shall goes beyond mere pattern recognition, emphasizing the imperative of comprehensive model construction.

Implementation of the Project:

1. The Dataset :

Omniglot is a dataset which is composed of 1623 different handwritten characters from 50 distinct alphabets. There are 30 alphabets for training (transfer learning) composed of 19.280 images, and 20 alphabets for one-shot classification task (unseen data) composed of 13.180 images. Each picture is 105 by 105 pixels. The Omniglot dataset was created to improve models that mimic human learning processes more closely. Each character was drawn by 20 individuals (independently from each other) via Amazon's Mechanical Turk platform. Each image of 50 alphabets is paired with stroke data in txt files, presenting sequences of [x,y,t] coordinates, time (t) is measured in milliseconds.

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2. Preprocessing :

Lake et al. (2015) employed a library of strokes to generate new types by choosing primitives. They did not mention how they created this library. However, this is probably the same library that they created at 2011.

Lake et al. (2011) created a library of pen strokes by employing txt files. They clustered 40,000 strokes by using k-means algorithm to form 1000 centroids. The library is composed of these 1000 stroke centroids. I am going to use the same preprocessing technique. I am also going to apply data augmentation techniques such as random cropping, rotation, and flipping to increase the diversity of the data. The 30 alphabets of the Omniglot dataset will be employed to pretrain a deep convolutional network and other 20 for 20-way one-shot classification task.

3. The Models Employed in Papers:

Lake et al. (2011) utilized a probabilistic model of character learning (the stroke model) which they introduced, alternatively from machine learning, the Deep Boltzmann Machine (DBM) which is also a generative probabilistic model and also the Nearest Neighbor (NN).

Lake et al. (2015) employed Bayesian Program Learning (BPL) and three deep learning models including deep convolutional networks. They used both image and stroke data from 30 alphabets (training set) for training BPL, and the image set was also utilized to pretrain the alternative deep learning models. Remaining 20 alphabets were used for classification.

I am going to use a deep convolutional network for 20-way one-shot classification task on Omniglot dataset, including implementation of *causality*, *compositionality*, and *learning-to-learn*.

4. 20-Way One-Shot Classification Task:

Lake et al. (2011) implemented 20 way classification from one example. Lake et al. (2015) also performed 20-way one-shot classification as a series of within-alphabet classification task for 10 different alphabets. They put forward that within-alphabet classification task is more challenging than the between-alphabet discrimination tasks used elsewhere. I am also going to implement 20-way one-shot within-alphabet classification.

5. Metrics to Validate the Model:

Lake et al, (2011) compared results from the stroke model (54.9 % correct), DBM (39.6 %) and NN (15.7 %) in pixels. Moreover, people were 97.6% correct and baseline was 75 %.

Lake et al., (2015) compared results from people (an average error rate of 4.5 %) (N=40), BPL (3.3 %), deep convolutional network (convnet; 13.5 %) and the HD model (34.8 %), and lastly a deep Siamese convolutional network (8.0 %), where chance is 95% errors. *The Modified Hausdorff Distance* was used as a baseline, producing 38.8 % errors. I am going to employ *The Modified Hausdorff Distance* as a baseline for 20-way one-shot classification.

References :

- 1 Lake, B. M., Salakhutdinov, R., & Tenenbaum, J. B. (2015). Human-level concept learning through probabilistic program induction. *Science*, 350(6266), 1332–1338. <https://doi.org/10.1126/science.aab3050>
- 2 Lake, B. M., Salakhutdinov, R., Gross, J., & Tenenbaum, J. B. (2011). One-shot learning of simple visual concepts. In Proceedings of the Annual Meeting of the Cognitive Science Society (CogSci).
- 3 Lake BM, Ullman TD, Tenenbaum JB, Gershman SJ. (2017). Building machines that learn and think like people. *Behavioral and Brain Sciences.*;40:e253. [doi:10.1017/S0140525X16001837](https://doi.org/10.1017/S0140525X16001837)